

2016.0 RANGE ROVER (LG), 413-13

PARKING AID

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the rear view camera system, refer to the relevant Description and Operation section of the workshop manual.

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Touch Screen (where fitted) ▪ Audio Head Unit (where fitted) ▪ Proximity Camera Control Module ▪ Rear View Camera	▪ Fuses ▪ Wiring harnesses and connectors ▪ Touch Screen (where fitted) ▪ Audio Head Unit (where fitted) ▪ Proximity Camera Control Module ▪ Rear View Camera

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes

(DTCs) and refer to the DTC Index

5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSE	ACTION
Rear view camera image slow to react	▪ System operation within specification	NOTE: After selecting reverse, it may take up to 20 seconds for the image to be displayed. ▪ No further action necessary
Blank screen	▪ Touch screen fault (where fitted) ▪ Audio head unit (where fitted) fault ▪ Rear view camera not functioning	▪ Using the manufacturer approved diagnostic system, check the touch screen for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the audio head unit for related DTCs and refer to the relevant DTC index ▪ GO to Pinpoint Test A.
Blue screen	▪ Video in signal absent	▪ GO to Pinpoint Test B.
No tracking lines	▪ Missing/invalid reverse gear signal ▪ LIN fault	▪ Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC index ▪ GO to Pinpoint Test C.
Frozen tracking lines	▪ LIN data gateway fault	▪ GO to Pinpoint Test D.

PINPOINT TESTS

NOTE:

A blank screen is the default display when the rear view camera is not transmitting an image.

CAUTION:

Do not probe the coaxial cable connectors as they are prone to damage.

NOTE:

A DC resistance measurement is not a reliable test method as the system operates at low voltage and high frequency.

PINPOINT TEST A : PERMANENT BLANK SCREEN TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: PERMANENT BLANK SCREEN TEST 1	
	1 Refer to the electrical circuit diagrams and check the power and ground connections to the rear view camera
	Are the power and ground circuits within specification? Yes GO to A2. No Repair power or ground circuit as necessary
A2: PERMANENT BLANK SCREEN TEST 2	
	1 Check the integrity of the rear view camera coaxial cable connectors (at rear view camera, the touch screen and in-line connectors)
	2 Check the coaxial cable for excessive bending, clamping and

	insulation damage
	Is the rear view camera coaxial cable disconnected or damaged? Yes Reconnect or install a new coaxial cable as necessary No Install a new rear view camera

NOTE:

A blue screen is the default display when the video in signal is absent.

PINPOINT TEST B : BLUE SCREEN TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: BLUE SCREEN TEST 1	
	1 Select reverse gear and observe the touch screen
	Is the touch screen blue? Yes Check the integrity of the rear view camera coaxial cable connectors (at rear view camera, the touch screen and in-line connectors), and retest No GO to Pinpoint Test A.

PINPOINT TEST C : ABSENT TRACKING LINES TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: ABSENT TRACKING LINES TEST 1	
	1 Refer to the electrical circuit diagrams and check the rear view camera LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
	Is a LIN bus circuit fault present? Yes Repair the LIN bus circuit as necessary No Install a new rear view camera

NOTE:

The vehicle may take 30 seconds (or 100m) to learn the steering centre position after starting the engine. This is normal.

PINPOINT TEST D : FROZEN TRACKING LINES TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: FROZEN TRACKING LINES TEST 1	
	1 Start the engine
	2 Wait at least 30 seconds
	3 Select reverse gear
	4 Turn the steering wheel and observe the touch screen
	Do the tracking lines react to steering input? Yes No fault present No Potential LIN data gateway fault

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - DTC: General Proximity Sensor Module \(GSPM\)](#) (100-00 General Information, Description and Operation) / [Diagnostic Trouble Code Index - DTC: Camera Control Module \(CCM\)](#) (100-00 General Information, Description and Operation) / [Diagnostic Trouble Code Index: Touchscreen](#) (100-00 General Information, Description and Operation).